THE RAVENCLAW

A DRONE WITH ROBOTIC ARM



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Are you looking for a stable aerial manipulator for completing high altitude precise tasks?
-We present to you "THE RAVENCLAW"

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Development and implementation of RavenClaw:

A stable aerial manipulator for high altitude manipulation tasks

1 Abstract:

What if a robotic arm can be mounted on a drone to enhance versatility and adaptability in various operational contexts! This paper presents the design, development, and practical implementation of our single-arm hexacopter drone, 'RAVENCLAW'. Our innovative approach focuses on the design and integration of a single arm equipped with multi-functional, customizable end-effectors, enabling a wide range of tasks to be performed efficiently. The integration of a robotic arm onto a drone platform enables a wide range of applications, including aerial manipulation, power line maintenance, infrastructure inspection, search and rescue operations, sample collection, environmental monitoring, and hazmat handling. The design process involved considerations such as payload capacity, power management, control system integration, and structural stability to ensure optimal performance and maneuverability. Additionally, the implementation of advanced sensing and computer vision technologies enables precise object detection, manipulation, and interaction capabilities for the drone-arm system. Experimental results demonstrate the effectiveness and versatility of the proposed platform in various real-world scenarios, showcasing its potential to revolutionize tasks traditionally limited to ground-based robots or human operators. This

research contributes to the advancement of UAV (Unmanned Aerial Vehicle) technology and opens up new avenues for autonomous aerial manipulation in diverse fields.

2 Description and Features:

The RavenClaw is a hexacopter drone with a robotic arm which combines the aerial mobility of a drone with the manipulation capabilities of a robotic arm, offering a versatile platform for a wide range of applications. It's able to perform aerial tasks which are difficult for a person to perform at high altitude. It consists of Hexacopter drone frame S550,1400KV BLDC motor, ESC(30A), 4200mAH 3s battery, Pixhawk 2.4.8 flight controller ,8045 Carbon fiber propeller (8 inch), Flysky FS-i6x (with FS-iA6B 2.4 GHz transmitter & receiver), Arm-frame (3D printed) with customizable gripper, MG996R servo motor, Arduino uno, Bluetooth module, Buck converter and other hardwares .



2.1 Key Features:

Here are some of the key features of our project:

a) 2 DOF arm with various end effectors: The robotic arm is equipped with multiple degrees of freedom, allowing it to articulate and maneuver in three-dimensional space. It typically consists of joints, links, and an end effector capable of grasping, lifting, and manipulating objects.

b) Precise Control Using the gripper: Advanced control algorithms enable precise positioning and manipulation of the robotic arm, ensuring accuracy in tasks such as object grasping, placement, or assembly.

c) Manipulate objects like wire, screws: Our end effectors were able to perform manipulating tasks like picking up specific wire, rotating a screw, pick up sharp objects using our gripper.

d) User friendly app to control arm: We developed an app named 'RavenClaws' which precisely controls the arm according to our need.

e) Stable flight while arm being operated: We were able to stably fly our drone while operating the arm.

f) Live HD POV from drone to user end: We are using a smartphone camera using 'DroidCam' software. Our drone has a smartphone holder. We take our input image from our phone and show the output in the application.

h)Modular Design: Customizable gripper and attachments for different tasks

g) Features of our drone:

Total weight	1.89 kg (with arm)
Drone weight	1.54 kg
Total thrust	5.4 kg (900g per motor)
Flight time	15 minutes
Range	500 m
Height	10m

3. Components, functions & description:

3,1 Component List

Component	Type & Ratings	Quantity	Function
Hardwares			
Frame	Hexacopter drone frame S550 (Carbon Fiber)	1	This lightweight carbon-fiber frame provides the base structural integrity.
Motor	EMAX 1400KV BLDC motor	6	The combination of 6 BLDC motors provide 5.4 kg thrust to the propellers which lifts our drone
ESC	ESC(30A)	6	The ESC controls the speed and direction of the motors
Battery	4200mAH 3s battery	1	The 4200mAH 3s battery provides the power to the motors & ESC
Power module	Battery to ESC power module with XT60 port	1	Converts the battery input voltage to 12V output and prevents current outflow while protecting the battery and motors.

Flight Controller	Pixhawk 2.4.8 flight controller	1	Our Pixhawk flight controller is the main component of our drone which is used for system calibration, flight stabilization, GPS, telemetry, and integration .
Propeller	8045 Carbon fiber propeller (8 inch)	3 pairs	The propellers are the key component of the drone to fly. We used a carbon fiber propeller so that it doesn't get electrocuted while working with electricity.
Transmitter & receiver	Flysky FS-i6x (with FS-iA6B 2.4 GHz transmitter & receiver)	1	Transmitter sends commands from the pilot to the radio frequency of the receiver of the drone. A drone is controlled by the transmitter
Arm frame	Arm-frame (3D printed) with customizable gripper	1	The arm connects the drone with the gripper
Servo Motor	MG996R servo motor	2	The servos provided the torque which enabled the rotation & movement of the arm & gripper
Arduino	Arduino uno	1	The arm control was completed using the arduino
Module	Bluetooth module	1	The arm & arduino was controlled by the app using a bluetooth module
Converter	Buck converter (to 5V)	1	The battery input was converted into 5V output to the arduino

Softwares:
Mission planner

DroidCam
RavenClaws



Hexacopter drone frame S550



EMAX 1400KV BLDC motor



ESC(30A)



4200mAH 3s battery



Power module with XT60 port



Pixhawk 2.4.8 flight controller



8045 Carbon fiber propeller



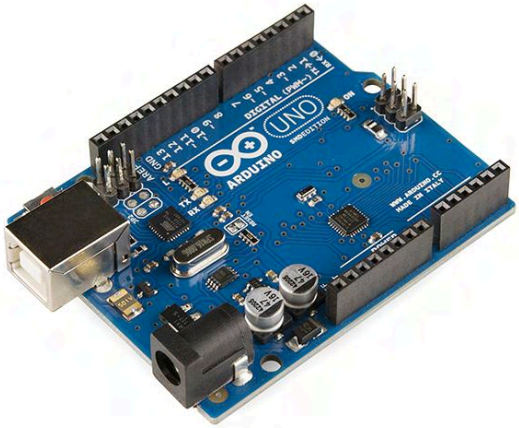
Flysky FS-i6x



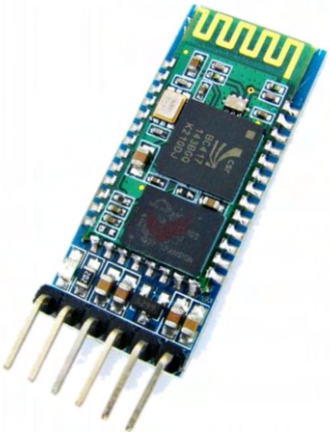
FS-iA6B 2.4 GHz receiver



MG996R servo motor



Arduino uno



Bluetooth module





Buck converter *Jumper wires*

Fig:3.1 List of components

3.2 Drone Frame



Fig 3.2.1: S550 Frame

The S550 Frame, a cornerstone of our drone's construction, boasts a robust design crafted from high-quality carbon fiber. Carbon fiber is renowned for its exceptional strength-to-weight ratio, making it an ideal choice for constructing durable and lightweight frames. This material ensures that the drone remains resilient in the face of various challenges, providing structural integrity even in adverse conditions.

Strength and Load Resistance: The S550 frame's inherent strength allows it to resist substantial loads, making it particularly adept at withstanding the impact forces resulting from crash landings or unexpected failures during flight. The carbon fiber construction disperses and absorbs energy efficiently, minimizing the risk of structural damage. This feature enhances the overall durability of the drone, contributing to its reliability in demanding operational environments.

Crash Landing and High-Altitude Resilience: In scenarios where the drone experiences crash landings or falls from a high altitude, the S550 frame's structural integrity becomes paramount. The carbon fiber composition provides the necessary strength to endure such incidents. The frame's ability to resist deformation and maintain its form ensures that critical components, such as flight controllers and sensors, remain protected, minimizing potential damage and facilitating easier maintenance and repairs.

Landing Gear Challenges: While the S550 frame itself offers exceptional durability, it is crucial to acknowledge certain challenges related to the landing gear. In our experience, the landing gear supplied with the S550 frame did not meet our expectations. The inadequacies in the landing gear design led to difficulties during landings and posed challenges in maintaining stability upon touchdown.

Issues Faced: The suboptimal landing gear design resulted in problems related to the drone's overall stability during landing operations. The issues included instability on uneven terrain, difficulty in absorbing impact forces during landings, and an increased risk of tipping over. These challenges impacted the overall performance and reliability of the drone in real-world applications.

Mitigations and Improvements: To address the landing gear challenges, we have explored alternative landing gear options and modifications that better align with our operational requirements. By seeking improvements in this crucial component, we aim to enhance the overall stability and reliability of the drone, ensuring a smoother landing experience in diverse terrains. In summary, the S550 frame, with its carbon fiber construction, stands out as a durable and resilient foundation for our drone. While the landing gear presented challenges, ongoing efforts to optimize this component will further enhance the drone's performance and usability in various operational scenarios.

We added this landing gear later on to improve the landing smoothness.

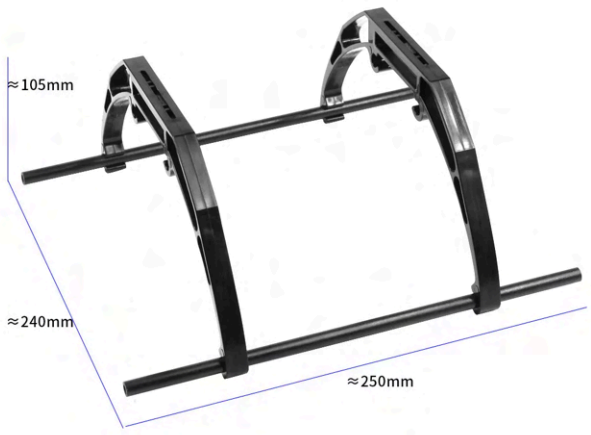


Fig 3.1.2: Improved Landing Gear

3.3 Flight Controller

3.3.1 Comparison of Different Flight Controllers

1. APM (ArduPilot Mega):

1. **Legacy Platform:** APM is an earlier generation flight controller that gained popularity for its open-source nature and compatibility with ArduPilot software.
2. **Processing Power:** APM controllers generally have lower processing power compared to Pixhawk, limiting their capabilities in handling advanced algorithms and features.
3. **Connectivity:** Limited connectivity options, with fewer ports for peripherals and sensors.

4. **Sensor Compatibility:** While APM is compatible with various sensors, its support for advanced sensors may be limited compared to Pixhawk.

2. Pixhawk:

1. **Modern and Advanced:** Pixhawk is a more modern and advanced flight controller, known for its increased processing power and feature-rich design.
2. **Processing Power:** Pixhawk controllers typically feature more powerful processors, enabling them to handle complex tasks and advanced algorithms with greater efficiency.
3. **Connectivity:** Pixhawk controllers offer a wider array of connectivity options, providing more ports for attaching peripherals and additional sensors.
4. **Sensor Compatibility:** Pixhawk supports a broad range of sensors, making it versatile and suitable for a variety of applications.

Advantages of Pixhawk Over APM:

Advanced Features: Pixhawk's increased processing power allows for the implementation of advanced flight modes, precise control algorithms, and sophisticated navigation features.

Connectivity: Pixhawk's enhanced connectivity options enable the integration of a diverse set of peripherals, such as GPS modules, telemetry systems, and additional sensors, expanding the drone's capabilities.

Sensor Support: Pixhawk's compatibility with a wide range of sensors, including a high-quality barometer, enhances its ability to gather accurate altitude and pressure data. This is particularly crucial for applications requiring precise altitude control.

Community and Development: Pixhawk has a large and active community of developers, contributing to ongoing updates, improvements, and support. This ensures that Pixhawk remains at the forefront of innovation in the field of open-source flight controllers.

While both APM and Pixhawk have their merits, Pixhawk emerges as the superior choice for those seeking a more modern, powerful, and feature-rich flight controller. The inclusion of a high-quality barometer enhances Pixhawk's capabilities in altitude-sensitive applications, making it a preferred option for a wide range of drone enthusiasts and professionals.

Challenges that we faced: (4-9)

4 Mission Planner

Calibrating a Pixhawk flight controller is a crucial step to ensure accurate sensor readings and optimal performance of your drone. Here's a general procedure for calibrating a Pixhawk:

Note: Before starting the calibration process, ensure that your Pixhawk is connected to a ground control station (GCS) software, such as Mission Planner or QGroundControl.

1. Connect Pixhawk to Your Computer
   Connect the Pixhawk flight controller to your computer using a USB cable.
   Power up the Pixhawk by connecting it to a battery.

2. Open Ground Control Station Software:
   Open the ground control station software on your computer (e.g., Mission Planner or QGroundControl).
   Connect to Pixhawk:

3. Establish a connection between your computer and the Pixhawk flight controller through the GCS software.
   Select the Correct COM Port

4. In the GCS software, navigate to the connection settings and ensure that the correct COM port is selected for communication with Pixhawk.

5. Access the Initial Setup Menu:
   In Mission Planner, go to the "Initial Setup" menu. In QGroundControl, navigate to the "Vehicle Setup" section.

6. Calibrate Accelerometer:
   Select the "Accelerometer" calibration option.
   Place the drone on a level surface and follow the on-screen instructions to calibrate the accelerometer. This step ensures accurate measurements of the drone's orientation.

7. Calibrate Compass:
   Choose the "Compass" calibration option.
   Follow the prompts to perform the compass calibration. Rotate the drone in multiple directions as instructed to calibrate the compass and account for magnetic interference.

8. Calibrate Gyroscope (Optional):

Some Pixhawk versions may have an option to calibrate the gyroscope. If available, select the "Gyroscope" calibration option and follow the on-screen instructions.

9. Calibrate Radio/Transmitter:
   If you are using a radio transmitter, navigate to the "Radio Calibration" section.
   Follow the prompts to move the control sticks through their full range of motion. This step ensures that the Pixhawk recognizes and correctly interprets transmitter inputs.

10. Save and Reboot:

After completing all calibration steps, save the settings and reboot the Pixhawk flight controller.
Check Calibration Status:

Verify that all calibration parameters show as "green" or "calibrated" in the GCS software. This ensures that the Pixhawk has successfully calibrated all sensors.

Perform Test Flights:
Conduct test flights in a controlled environment to ensure that the drone responds correctly to user inputs and maintains stable flight.
By following these steps, you can effectively calibrate your Pixhawk flight controller, ensuring accurate sensor readings and optimal performance during flight operations. Calibration procedures may vary slightly depending on the specific GCS software and Pixhawk firmware version you are using, so refer to the documentation corresponding to your setup for additional guidance.

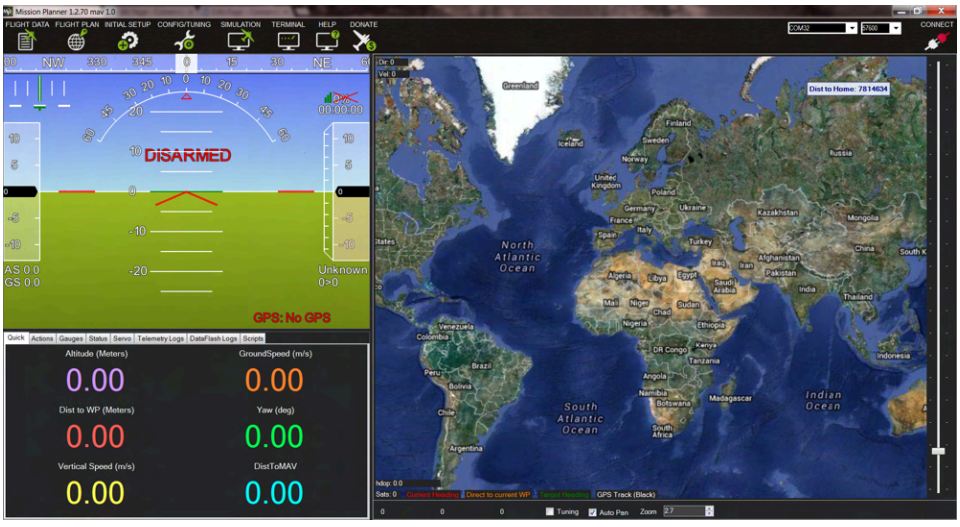


Fig 4.1. Mission Planner Interface

5 CAD Design

5.1 Main Arm Body Design (Initial)

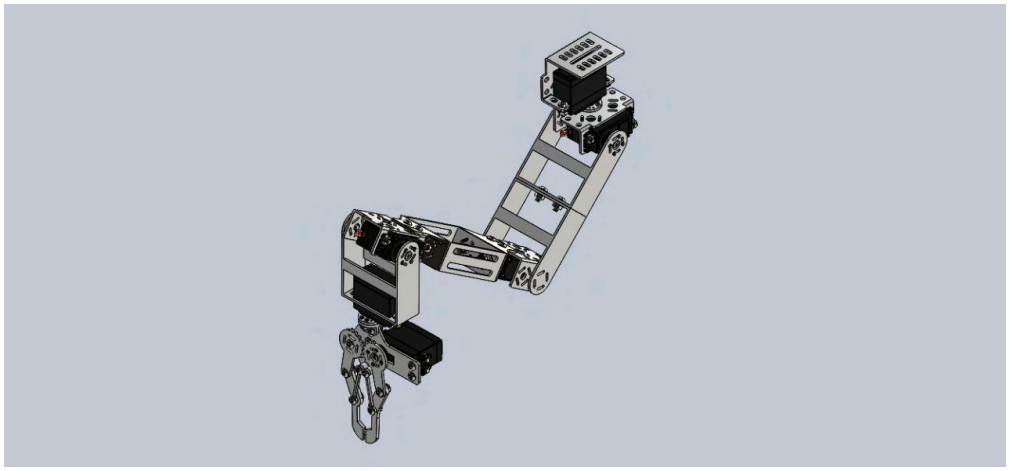
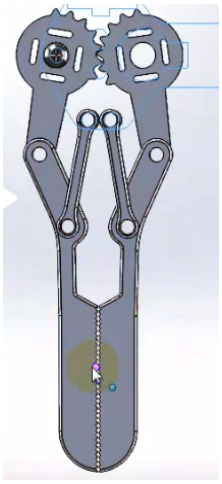


Fig 4. Proposed Arm

We developed this 6 DOF Arm for our drone which was to be mounted in the gimbal frame of the drone. But it led to some issues for our drone. Our drone wasn't being stable with this much of freedom.

5.2 Gripper Design

We designed a gripper for gripping wire. One of the features of our drone is to ensure the accessibility to customized grippers. This was one of them. It was made to grab long and thick wires.

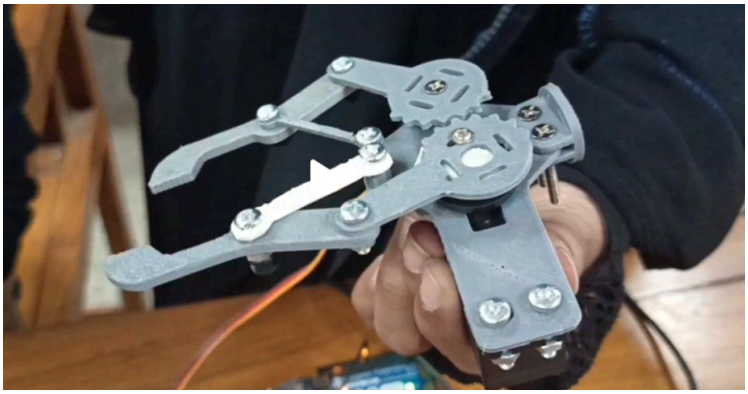


6 3D Printing

One of the crucial parts of our project was to 3D print the designs we created. This part was tough. We couldn't manufacture things like we wanted to.



As we can see one of the connectors broke so we had to use a PVC board cutting to fill the gap before we get another printing of the gripper.



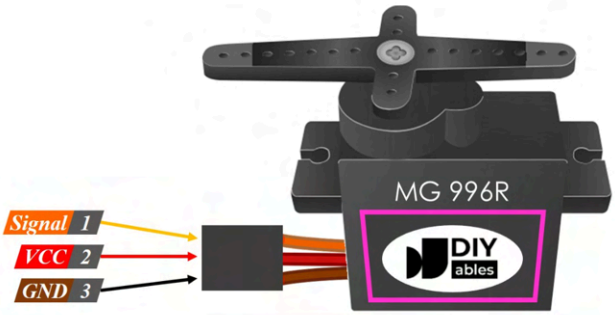
As many parts were not printed properly, we had to assemble the arm with PVC board parts.



7 Control of Arm

To control the arm we used MG996R servo motors. Since it has more torque than other available motors. But in reality it doesn't.

7.1 MG996R is a scam!





In Bangladesh, the MG996R servo, while widely available, has been reported to exhibit reduced torque compared to the expected specifications. The MG996R is known for its high torque output, but inconsistencies in manufacturing or variations in product quality may result in some units deviating from the specified torque values. Users in Bangladesh have experienced instances where the actual torque delivered by the MG996R servo falls short of anticipated levels. It is essential for consumers to be aware of these potential variations in performance and consider alternative options or sources if a higher torque is a critical requirement for their specific applications. Thorough testing and verification of individual servo units may be advisable to ensure compatibility with the intended use cases in the local market.

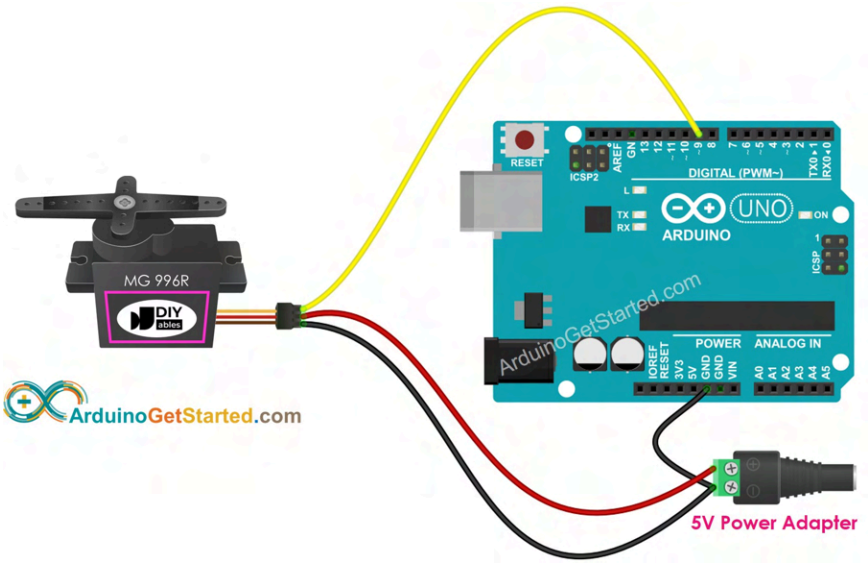
As MG996R couldn't carry enough loads. The first three or four motors broke down.

This is one of the reasons why we reduced the DOF of our arms. [Discussed later on the documentation]

7.2 Arduino and source code

This is for testing one servo. Full Arm control with bluetooth module is provided later on.

Schematic of Arduino Connections:



Code:

```
#include <Servo.h>

Servo servo;  // create servo object to control a servo

void setup() {
  servo.attach(9);  // attaches the servo on pin 9 to the servo objectu
  servo.write(0);   // rotate slowly servo to 0 degrees immediately
}

void loop() {
  for (int angle = 0; angle <= 180; angle += 1) {  // rotate slowly from 0 degrees to 180 degrees, one by one degree
    // in steps of 1 degree
    servo.write(angle);  // control servo to go to position in variable 'angle'
    delay(10);         // waits 10ms for the servo to reach the position
  }

  for (int angle = 180; angle >= 0; angle -= 1) {  // rotate from 180 degrees to 0 degrees, one by one degree
    servo.write(angle);                          // control servo to go to position in variable 'angle'
    delay(10);                                  // waits 10ms for the servo to reach the position
  }
}
```

7.3 Arm Control with Bluetooth Module

Code:

```
#include <Servo.h>
Servo baseServo;
Servo linkServo;
Servo gripperServo;
const int BASE_SERVO_PIN = 8;
const int LINK_SERVO_PIN = 10;
const int GRIPPER_SERVO_PIN = 9;
void setup() {
 Serial.begin(9600);

 linkServo.attach(LINK_SERVO_PIN);
 gripperServo.attach(GRIPPER_SERVO_PIN);
 gripperServo.write(90);
 linkServo.write(180);
}
void gripper_on_off(){
  gripperServo.write(90);
  delay(1000);
  gripperServo.write(30);
  delay(1000);
}
void gripper_on(){
  gripperServo.write(90);
}
void gripper_off(){
 gripperServo.write(30);
}
void gradually_gripper_on(){
  gripperServo.write(gripperServo.read() + 2);
}
void gradually_gripper_off(){
  gripperServo.write(gripperServo.read() - 2);
}
void loop() {
 if(Serial.available()) {
   char c = Serial.read();
   Serial.println(c);
   if(c == 'l'){
```

```
    linkServo.write(linkServo.read() + 2);
  }
  if(c == 'r'){
    linkServo.write(linkServo.read() - 2);
  }
  if(c == 'k') {
    gripperServo.write(gripperServo.read() + 5);
  }
  if(c == 'c') {
    gripperServo.write(gripperServo.read() - 5);
  }
  if(c == 'w') {
    gradually_gripper_on();
  }
  if(c == 'x') {
    gradually_gripper_off();
  }
 }
}
```

8 Truncation of DOF in Arm

Reducing the Degrees of Freedom (DOF) from 6 to 1 in the robotic arm of a drone signifies a significant simplification in the mechanical structure and control system. A typical robotic arm with 6 DOF is designed to move freely in three-dimensional space, allowing for complex and versatile motions. However, in the context of a drone, this high degree of freedom can introduce challenges related to stability, control complexity, and weight distribution.

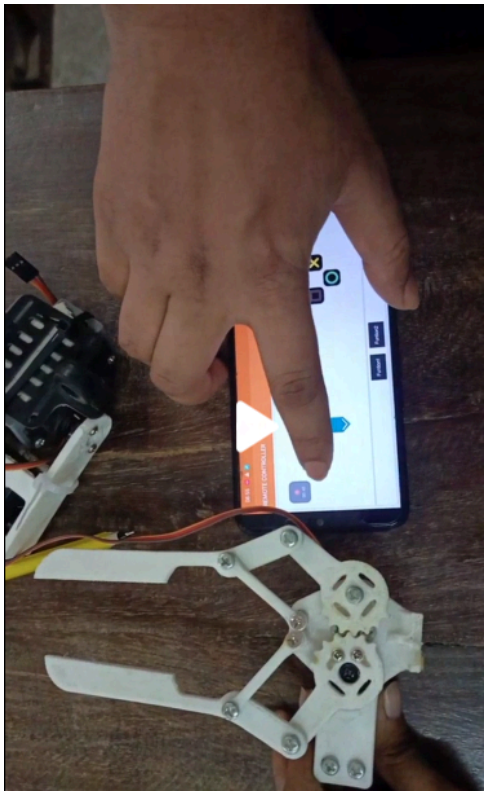
Stability Improvement:
A drone's stability is crucial for safe and controlled flight. A 6-DOF robotic arm introduces additional variables that can potentially affect the drone's center of gravity and overall balance. By reducing the DOF to 1, the system becomes inherently more stable, as it is now constrained to a single axis of motion. This simplification can lead to better predictability and control over the drone's behavior during flight, particularly in dynamic and unpredictable environments.

Control System Complexity:
A higher number of DOF in a robotic arm necessitates a more intricate control system to manage and coordinate the movement of each joint. Simplifying the arm to 1 DOF greatly reduces the complexity of the control algorithms, making it easier to implement and fine-tune. This reduction in complexity not only streamlines the control system but also minimizes the computational load on the drone's onboard processors, contributing to more efficient overall performance.

Weight and Payload Considerations:
A multi-DOF robotic arm typically adds weight to the drone, which can impact its flight characteristics, agility, and energy consumption. By reducing the DOF, the weight of the arm is reduced, potentially allowing for better weight distribution and improved flight efficiency. This is particularly important for drones with limited payload capacity, where minimizing additional weight is essential for maximizing flight time and operational capabilities.

Specific Application Focus:
Reducing the robotic arm's DOF suggests a shift towards a specific application or task that requires a simpler and more focused range of motion. Depending on the intended use, such as picking up objects, performing specific maneuvers, or interacting with the environment in a controlled manner, a 1-DOF arm may offer the necessary functionality without introducing unnecessary complexity.

In summary, the decision to reduce the DOF from 6 to 1 in the robotic arm of a drone reflects a strategic choice to enhance stability, simplify control systems, optimize weight distribution, and tailor the drone's capabilities for specific applications. This trade-off aligns with the need to balance the intricacies of a multi-DOF robotic arm with the practical requirements of drone flight, ensuring a more stable and efficient platform for targeted tasks.



We used the movement of the gripper and its rotation only.

9 Sabilization of Drone

Stabilizing a drone with moving arms, particularly due to the pilot's expertise, involves a combination of skillful manual control, understanding the drone's dynamics, and making real-time adjustments to counteract any destabilizing factors. Here is a general procedure highlighting key steps taken by **an expert pilot**:

Thorough System Understanding:

The pilot must have a comprehensive understanding of the drone's design, including the dynamics of the moving arms. This involves awareness of the drone's center of gravity, weight distribution, and the impact of arm movements on stability.

Calibration and Configuration:

Ensure that the drone's sensors, including accelerometers and gyros, are properly calibrated. Accurate sensor readings are essential for the pilot to make informed decisions during flight.

Expert Manual Control:

Rely on the pilot's expertise to manually control the drone. Skilled pilots can intuitively counteract destabilizing factors, such as arm movements, by making precise adjustments to the drone's orientation and thrust.

Real-Time Monitoring:

Continuously monitor the drone's behavior in real-time. Expert pilots develop a keen sense of how the drone responds to different inputs, allowing them to anticipate and counteract potential instabilities caused by moving arms.

Swift Reactions to Arm Movements:

As the drone's arms move, the pilot must react swiftly to any changes in stability. This involves making instantaneous adjustments to the drone's pitch, roll, and yaw to maintain a balanced and controlled flight.

Adaptive Flying Techniques:

Employ adaptive flying techniques that take into account the specific dynamics introduced by the moving arms. This may include adjusting the drone's orientation, speed, or altitude based on the arm's position to mitigate destabilizing effects.

Fine-Tuning Control Inputs:

Expert pilots possess the ability to provide nuanced control inputs. Fine-tuning the control stick movements allows for precise adjustments, enabling the pilot to counteract any unintended movements resulting from the drone's arm motion.

Practice and Experience:

Achieving proficiency in stabilizing a drone with moving arms requires practice and experience. Expert pilots have likely spent extensive time honing their skills through simulations and actual flights, developing an innate understanding of how the drone behaves in various scenarios.

Emergency Response Preparedness:

In the event of unexpected issues arising from arm movements, expert pilots are prepared to execute emergency maneuvers. This may involve activating emergency stabilization modes or utilizing fail-safe procedures to regain control.

Continuous Training and Skill Enhancement:

Stabilizing a drone with moving arms is an ongoing learning process. Expert pilots engage in continuous training, stay updated on advancements in drone technology, and refine their skills to adapt to new challenges.

By combining these elements, a skilled pilot can successfully stabilize a drone with moving arms, showcasing the importance of expertise, experience, and adaptive control strategies in managing dynamic flight scenarios.



Also we have achieved stability at different heights:



10 Image Transfer from Drone to Ground Station

To achieve this we used droidcam software which was installed in both Phone and PC. Phone was used as a camera which transmitted images to the PC.

DroidCam is an Android application that allows you to use your smartphone as a webcam for your computer. It is commonly used for video conferencing, online meetings, or live streaming. Here's a step-by-step guide on how to use DroidCam:

Note: DroidCam is available in both a free and a paid version. The paid version provides additional features and higher resolution.

Using DroidCam:

1. Download and Install:

   Download and install the DroidCam app on your Android device from the Google Play Store.
   Download DroidCam Client on PC:

   Visit the DroidCam website (https://www.dev47apps.com/) on your computer.
   Download and install the DroidCam Client application suitable for your operating system (Windows or Linux).

2. Connect Your Android Device:

   Open the DroidCam app on your Android device.
   Make sure your Android device is connected to the same Wi-Fi network as your computer.
   Take note of the Device IP and DroidCam Port displayed on the app's main screen.

3. Launch DroidCam Client on PC:

   Open the DroidCam Client application on your computer.

4. Configure DroidCam Client:

   In the DroidCam Client on your computer, enter the Device IP and DroidCam Port obtained from the Android app.

Choose the preferred video and audio settings such as resolution and FPS.

5. Connect:

   Click the "Start" button in the DroidCam Client on your computer.
   The DroidCam Client will attempt to connect to your Android device using the provided IP and port.

6. Activate Webcam:

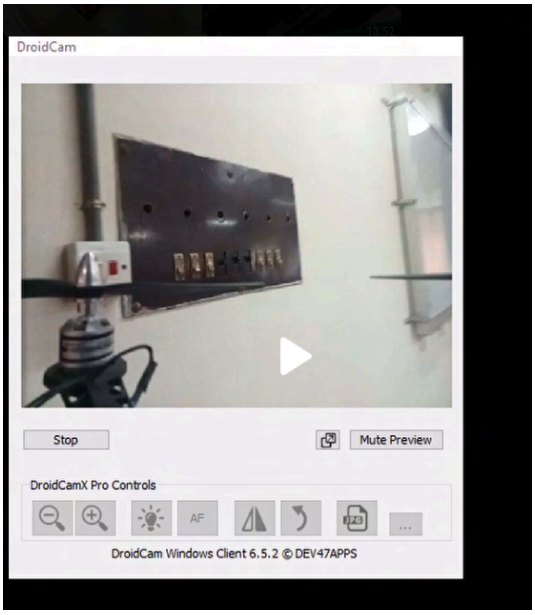
   Once the connection is established, your Android device's camera will be activated as a webcam.
   You can now use the smartphone camera in applications like Zoom, Skype, OBS Studio, or any other software that allows you to select a webcam.

To stop using DroidCam, click the "Stop" button on the DroidCam Client on your computer.

Close the DroidCam app on your Android device.

Tips:

1. Ensure both your computer and Android device are on the same Wi-Fi network.
2. If you face connection issues, try restarting both the DroidCam app on your Android device and the DroidCam Client on your computer.
3. In applications like Zoom or Skype, select "DroidCam" as the webcam source.
4. By following these steps, you should be able to effectively use DroidCam to turn your Android device into a webcam for your computer.



11 Picture of our project: Proposal to Reality

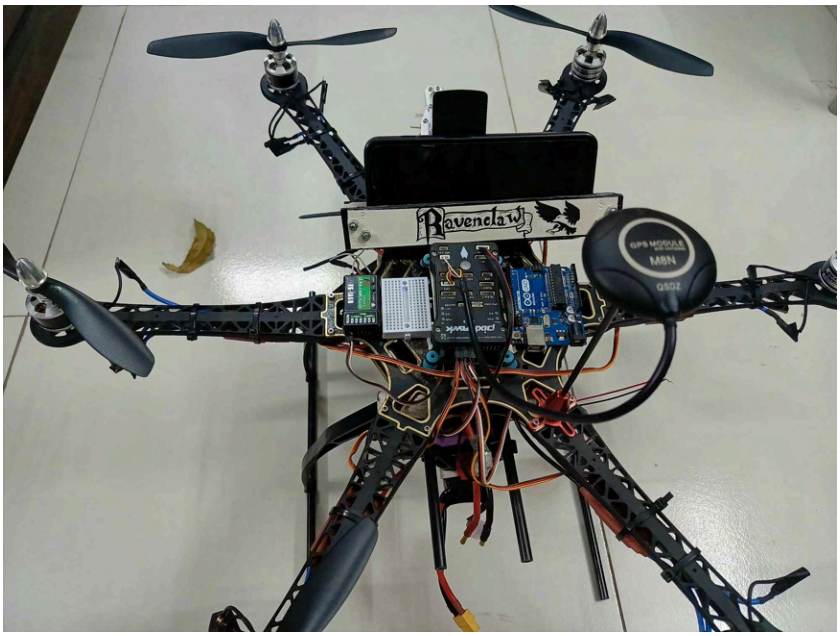
Our proposed Drone was:



Reality:






Ravenclaw
GPS MODULE
M8N


Ravenclaw





12 Application

12.1 Application of Aerial Manipulator

Maintenance and Inspection:

Infrastructure Inspection: Aerial manipulators can be used to inspect and maintain infrastructure such as bridges, towers, and pipelines. They can open and adjust screws for routine maintenance or repairs.

Power Line Maintenance: Drones with grippers can be employed to tighten or loosen bolts and screws on power lines, reducing the need for human workers to climb utility poles.

Construction and Assembly:

Building Construction: Aerial manipulators can assist in construction tasks by gripping and placing building materials or adjusting components during assembly.

Assembly Line Assistance: In manufacturing, drones with grippers can be programmed to perform specific tasks on assembly lines, such as tightening screws or adjusting components.

Emergency Response:

Search and Rescue: Drones equipped with grippers can be used in search and rescue missions to grab and lift objects, clear debris, or open access points to reach trapped individuals.

Disaster Response: Aerial manipulators can aid in disaster-stricken areas by adjusting or fixing damaged structures, or by retrieving important objects for rescue operations.

Environmental Monitoring:

Ecological Sampling: Drones can be employed to collect environmental samples by grabbing plant specimens or soil samples for research purposes.

Wildlife Conservation: Aerial manipulators can be used to handle sensors or equipment in wildlife studies without disturbing the animals.

Warehouse and Logistics:

Inventory Management: Drones with grippers can be used to pick and place items in warehouses, facilitating efficient inventory management.

Package Handling: Aerial manipulators can assist in loading and unloading cargo by grabbing and securing packages.

Precision Agriculture:

Crop Management: Drones can be equipped with grippers to perform tasks like adjusting sensors, securing small devices, or collecting samples for precision agriculture applications.

Medical Applications:

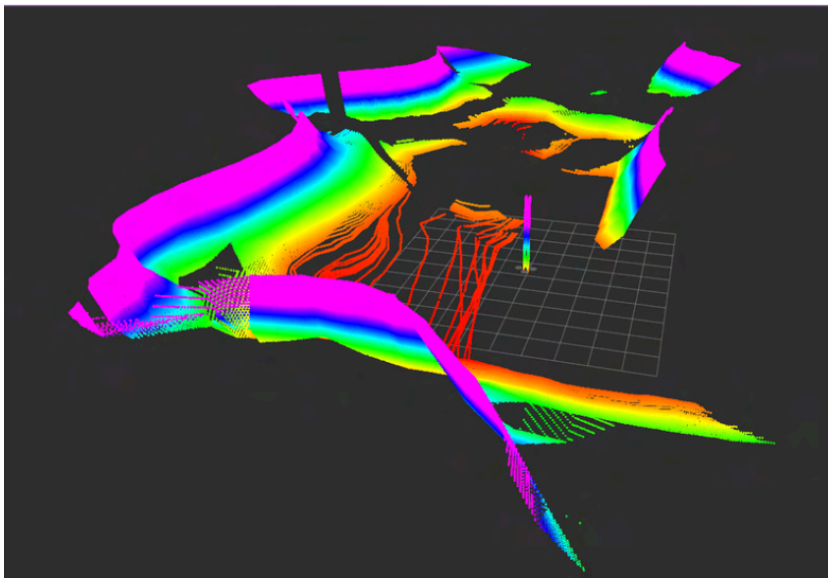
Medical Deliveries: Aerial manipulators can be used to transport medical supplies or assist in medical procedures by carrying and manipulating necessary tools or equipment.
Security and Defense:

Surveillance and Inspection:

Drones with manipulators can inspect and manipulate objects in security and defense contexts, such as checking and adjusting surveillance equipment.

13 Future Recommendation:

- ➢ Increasing the arm's degree of freedom and length
- ➢ Integrating different sensors for more inspection
- ➢ Increasing payload
- ➢ Making the flight autonomous.



Implementing autonomous features for advance application. (Work Ongoing)

14 Conclusion:

Our exploration of the single-arm drone design has revealed its immense potential for revolutionizing various industries. As we continue to push the boundaries of innovation, the single-arm drone stands as a testament to the limitless possibilities of unmanned aerial vehicles in shaping the future of autonomous flight

15 Reference:

.1. https://ardupilot.org/planner/docs/common-mission-planning.html

2. https://www.ros.org/

3. https://projecthub.arduino.cc/

4. https://www.flysky-cn.com/